

Commerce XI Maths App

Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 2 - Functions | Total questions: 7

Question 1 - MCQ · PG-31 · LR-1

A function $f: A \rightarrow B$ associates every element of A with

Options

- A. a unique element of B
- B. two distinct elements of B
- C. every element of B
- D. no element of B

Correct answer: a unique element of B

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is a unique element of B.

Question 2 - MCQ · PG-31 · LR-2

If $(x, y) \in f$, then y and x are called respectively

Options

- A. image and pre-image
- B. pre-image and image
- C. domain and range
- D. co-domain and image

Correct answer: image and pre-image

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is image and pre-image.

Question 3 - MCQ · PG-31 · LR-3

A function $f: A \rightarrow B$ is one-one if

Options

- A. distinct elements in A have distinct images in B
- B. every element of B has a pre-image
- C. all elements of A have one common image
- D. $A = B$

Correct answer: distinct elements in A have distinct images in B

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is distinct elements in A have distinct images in B.

Question 4 - MCQ · PG-31 · LR-4

A function $f: A \rightarrow B$ is onto if

Options

- A. every element of B is an image of some element of A
- B. some element of B has no pre-image
- C. every element of A has two images
- D. the range is empty

Correct answer: every element of B is an image of some element of A

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is every element of B is an image of some element of A.

Question 5 - MCQ · PG-31 · LR-5

A function is into if

Options

- A. at least one element of B has no pre-image in A
- B. every element of B has a pre-image
- C. every element of A has no image
- D. it is one-one and onto

Correct answer: at least one element of B has no pre-image in A

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is at least one element of B has no pre-image in A.

Question 6 - MCQ · PG-31 · LR-6

If $f: A \rightarrow B$ and $g: B \rightarrow C$, the composite function from A to C is denoted by

Options

- A. $g \circ f$
- B. $f \circ g$ only
- C. $f + g$
- D. fg

Correct answer: $g \circ f$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $g \circ f$.

Question 7 - MCQ · PG-31 · LR-7

For an invertible function f , which statement is correct?

Options

- A. Domain of f = Range of f^{-1}
- B. Domain of f = Domain of f^{-1} always
- C. Range of f is empty
- D. f^{-1} does not interchange domain and range

Correct answer: Domain of f = Range of f^{-1}

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is Domain of f = Range of f^{-1} .